

Fat body transcriptional responses to dietary macronutrient balance and *Metarhizium* infection for desert locusts, *Schistocerca gregaria*

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Highlights

Note: this will be included as a separate file in the submission, but putting it here so that it's easy for everyone to draft and comment on. Each bullet points has to be 100 characters or fewer, with spaces.

- Fungal infection strongly reshaped the desert locust fat body transcriptome
- Infection-responsive expression was broadly shared across nutritional states
- A core of 467 infection-responsive genes was shared among all three diets
- Formal interaction tests supported selective, count-sensitive diet effects

Graphical abstract

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Abstract

Dietary macronutrient balance alters disease outcome in desert locusts, yet carbohydrate-biased nutrition can improve survival despite weaker phenoloxidase responses. We tested two competing explanations for this physiological paradox: enhanced immune investment under carbohydrate-biased nutrition versus diet-dependent modification of the host metabolic environment during infection. Fat body transcriptomes were measured from 44 fifth-instar female *Schistocerca gregaria* assigned to carbohydrate-biased (diet 33), balanced (diet 50), or protein-biased (diet 83) diets and challenged with *Metarhizium robertsii* or an oil control. Infection altered 1,016, 2,009, and 1,030 genes within diets 33, 50, and 83, respectively, with 467 genes shared among all three responses. Immune recognition, extracellular defense, transport, and Toll/Imd-associated functions were induced across nutritional conditions, whereas cuticle, cell-cycle, and selected metabolic functions generally decreased. A formal diet-by-infection likelihood-ratio test identified only nine genes in the primary transcript-plus-exon analysis and none under exon-only counting, indicating limited and count-definition-sensitive interaction. Post hoc curated immune candidates were commonly induced across diets, whereas protein-anabolism, nutrient sensing, amino-acid metabolism, and storage candidates showed heterogeneous responses. These results do not support generalized enhancement of immune transcription as the explanation for carbohydrate-associated survival. Instead, they are consistent with selective effects of nutritional state on host metabolism, resource availability, or tolerance, while leaving the causal mechanism unresolved.

73

74 **1. Introduction**

75 Nutrition and immunity are fundamental determinants of organismal fitness because they
 76 jointly influence health, survival, and disease susceptibility (Ponton et al., 2020). Nutritional
 77 resources are essential for the development and maintenance of immune defenses, which protect
 78 organisms against the continual threat of pathogenic infection (Schmid-Hempel, 2021). At the
 79 same time, the activation and maintenance of immune responses are energetically demanding
 80 processes that require substantial metabolic investment (Cotter et al., 2019). Consequently,
 81 nutrient acquisition, energy metabolism, and immune defense have evolved as highly integrated
 82 physiological processes (Ponton et al., 2023). Systems involved in nutrient sensing and
 83 utilization share signaling pathways and regulatory networks with those responsible for pathogen
 84 detection and immune activation, resulting in extensive crosstalk among metabolic, endocrine,
 85 and immune functions (Iyer et al., 2015). This interconnectedness forms the basis of
 86 immunometabolism and highlights how nutritional resources can influence immune
 87 performance, host–pathogen interactions, and disease outcomes (Pan et al., 2018). Although
 88 nutritional state is increasingly recognized as a major determinant of pathogen resistance and
 89 infection outcomes, the molecular mechanisms through which dietary macronutrient balance
 90 influences immune and metabolic responses to infection remain poorly understood (Ayres &
 91 Schneider, 2009; Ponton et al., 2023).

92 Numerous studies in insects demonstrate that the effects of food intake on host-pathogen
 93 interactions are highly context dependent, varying with the infecting pathogen, and host species.
 94 For example, in *Drosophila*, infection-induced anorexia increased tolerance to *Salmonella*
 95 *typhimurium* while decreasing resistance to *Listeria monocytogenes* (Ayres & Schneider, 2009).
 96 In the silkworm, *Bombyx mori*, dietary restriction through reduced food intake accelerated
 97 melanization, reduced early *Staphylococcus aureus* burdens, delayed mortality, and altered
 98 immune-related gene expression (Pan et al., 2018). Together, these studies demonstrate that total
 99 nutrient intake can influence infection outcomes through effects on pathogen burden, and host
 100 immune responses.

101 Beyond overall food intake, the balance of dietary macronutrients has emerged as an
 102 important determinant of infection outcomes. Over the past decades, the development of the
 103 geometric framework for nutrition by (Simpson & Raubenheimer, 2012) has substantially
 104 advanced our understanding of how macronutrient balance regulates animal physiology. This
 105 framework has enabled numerous studies examining how variation in dietary protein,
 106 carbohydrate, and lipid composition influences traits such as lifespan, reproduction, metabolism,
 107 and immune function across diverse animal systems (Ponton et al., 2023). Despite these
 108 advances, the extent to which macronutrient balance alters transcriptional responses to infection
 109 and ultimately shapes susceptibility to pathogenic challenge remains poorly understood. Existing
 110 studies suggest that the relationship between nutrition and pathogen resistance varies
 111 substantially among host species and infection types (Ponton et al., 2023). For example, In
 112 *Drosophila melanogaster*, protein limitation did not affect resistance (the ability to reduce

pathogen burden) to either *Escherichia coli* or *Lactococcus lactis*, but reduced tolerance (the ability to reduce the fitness costs of infection) to *E. coli*, highlighting that nutritional effects on host defense are pathogen dependent. (Kutzer & Armitage, 2016). Similarly, female *Drosophila* consuming carbohydrate-biased diets exhibited increased antimicrobial protein production and improved survival when infected with the bacteria *Micrococcus luteus* (Ponton et al., 2020). Carbohydrate-biased diets also increased longevity in bumblebees infected with *Nosema ceranae* despite increasing infection prevalence, indicating that improved survival does not necessarily coincide with reduced pathogen burden (Gómez-Moracho et al., 2021). Conversely, in honey bees coinfecting with two pathogens (*Nosema ceranae* and deformed wing virus B), protein supplementation reduced viral loads and improved survival, despite also promoting greater fungal *Nosema* replication (Tritschler et al., 2017). For grasshoppers and locusts infected with *Metarhizium* fungi, there are also some species specific differences, though diets high in carbohydrates and low in protein are generally protective for the host (Graham et al., 2014; Tahir et al., 2026; Zembrzuski et al., 2023). Supporting a potential mechanistic basis for these patterns, De Fine Licht et al. (2025) found that *Metarhizium* growth is primarily limited by protein availability and that locust (*Locusta migratoria*) tissues differ markedly in protein and carbohydrate composition, indicating that variation in host nutrient availability may directly shape fungal growth and development. Collectively, these studies indicate that dietary macronutrients do not uniformly enhance or suppress host defense; instead, infection outcomes emerge from interactions among host nutritional states, pathogen nutritional requirements, immune regulation, and the particular metric of disease being evaluated, including pathogen burden, survival, resistance, or tolerance.

One potential explanation for these contrasting interactions between diet and insect responses to pathogens lies in the intimate interactions between metabolic pathways (acquisition, storage, and allocation of nutrients) and immune signaling pathways. Although these systems were historically studied independently, increasing evidence indicates that they are tightly interconnected and evolutionarily conserved from insects to vertebrates, giving rise to the field of immunometabolism (K.-A. Lee & Lee, 2018). In insects, immunity against fungi and Gram-positive bacteria is primarily mediated by the Toll-Dif/Dorsal pathway, whereas immunity against Gram-negative bacteria is controlled through the Imd-Relish pathway, both of which culminate in AMP production (Galenza & Foley, 2019). The JAK/STAT pathway also contributes to immune defense, particularly against viral pathogens, and exhibits functional interactions with IIS signaling (Mukherjee et al., 2025). Current models propose that infection induces a coordinated shift away from anabolic growth and nutrient storage toward defense-oriented metabolic states. In *Drosophila*, activation of the Toll immune pathway suppresses insulin signaling, leading to reduced growth and nutrient storage as resources are redirected toward immune defense. Complementing this response, nutrient-sensing pathways such as TOR and FOXO interact with innate immunity, with poor diets reducing TOR signaling, increasing FOXO-activation, while increasing antimicrobial peptide expression and enhancing immune function (DiAngelo et al., 2009; Ponton et al., 2020). These signaling interactions can reallocate energetic resources from growth and storage toward immune defense, illustrating how metabolic and immune pathways are integrated during infection (K.-A. Lee & Lee, 2018). Together, these

155 studies illustrate how nutritional status is translated into immune responses through crosstalk
 156 between conserved metabolic regulators and canonical immune signaling pathways (Kim et al.,
 157 2022; Mukherjee et al., 2025b; Ponton et al., 2020).

158 Understanding how nutrition influences immune responses requires examining the
 159 molecular mechanisms that underlie these interactions. Transcriptomic analyses provide a
 160 powerful approach for investigating changes in gene expression in response to environmental
 161 and physiological factors (Mierziak et al., 2021). Diet, physical activity, pharmaceuticals, and
 162 disease can all alter patterns of gene expression, influencing the risk of pathological changes
 163 within an organism (Panagiotou & Nielsen, 2009; Phillips, 2013). Comparisons of diet-
 164 dependent transcriptomic profiles among healthy and diseased populations have therefore
 165 become increasingly valuable for identifying molecular biomarkers and improving our
 166 understanding of how nutritional factors influence health and disease processes (Trujillo et al.,
 167 2006).

168 In this study, we used the desert locust, *Schistocerca gregaria*, as a model to investigate
 169 how dietary macronutrient balance (protein-to-carbohydrate ratio) influences molecular
 170 responses to infection by the entomopathogenic fungus *Metarhizium robertsii*. For grasshoppers
 171 and locusts infected with *Metarhizium* fungi, while there are some species specific differences,
 172 diets high in carbohydrates and low in protein are generally protective for the host. However, the
 173 mechanisms of this protection are unclear (Graham et al., 2014; Tahir et al., 2026; Zembrzuski et
 174 al., 2023). We recently demonstrated that desert locusts consuming a carbohydrate-biased diet
 175 survived fungal infection longer than individuals fed balanced or protein-biased diets, despite
 176 exhibiting lower levels of some aspects of immune activation (Tahir et al. 2026). Specifically,
 177 locusts maintained on balanced and protein-biased diets displayed elevated phenoloxidase (PO)
 178 activity relative to those consuming a carbohydrate-biased diet. However, insect immunity is
 179 complex and many-faceted, using both cellular and humoral responses (Zdybicka-Barabas et al.,
 180 2025). Cellular defenses are mediated by hemocytes, which carry out phagocytosis, nodulation,
 181 and encapsulation, whereas humoral immunity involves activation of the phenoloxidase cascade
 182 that drives melanization, induction of immune signaling pathways such as Toll, Imd, and
 183 JAK/STAT to stimulate antimicrobial peptide and other immune effector production, and
 184 hemolymph coagulation to facilitate wound repair (Cammarata-Mouchtouris et al., 2022;
 185 Kordaczuk, 2024; Muhammad et al., 2024; Stączek et al., 2023; Valanne et al., 2022; Wojda et
 186 al., 2020). The observation that dietary macronutrient balance influenced phenoloxidase activity
 187 but not hemocyte abundance in *Metarhizium*-infected locusts suggests that nutritional effects on
 188 immunity may be complex and pathway-specific.

189 To more comprehensively understand how diet and *Metarhizium* infection affect locust
 190 physiology, we compared transcriptomic responses in the fat bodies of infected and uninfected
 191 locusts maintained on carbohydrate-biased, balanced, or protein-biased diets. We elected to focus
 192 on the fat body as this is a multifunctional metabolic and immune organ that serves as a major
 193 site of nutrient storage and metabolism while also synthesizing many of the immune proteins
 194 released into the hemolymph during infection (Arrese & Soulages, 2010; Sawa et al., 2025). We
 195 hypothesized that dietary macronutrient balance alters host responses to fungal infection through

nutrient-sensitive signaling pathways and changes in resource allocation between immune defense and anabolic metabolism. As prior studies have shown that high protein diets promote growth (K. P. Lee et al., 2006; Longo & Anderson, 2022; Tahir et al., 2026) we predicted that protein-biased diets would increase transcription of genes involved in growth and anabolic processes while possibly suppressing gene transcription of some aspects of immune-responses, and further predicted that carbohydrate-biased diets would reduce transcription of growth and anabolic genes while increasing transcription of genes involved in immune pathways.

2. Materials and methods

2.1 Experimental design

We used a 3 x 2 factorial design comprising three isocaloric diets and two inoculation treatments. The diets contained 33%, 50%, or 83% protein as a proportion of total dietary macronutrient, corresponding to whole-diet protein:carbohydrate percentages of 14:28, 21:21, and 35:7, respectively. Individual fifth-instar female *Schistocerca gregaria* were randomly assigned to a diet and received either *Metarhizium robertsii* conidia in oil or sterile oil alone. The RNA-seq data set analyzed here comprised 44 individuals: diet 33 included seven controls and nine infected locusts, diet 50 included six controls and nine infected locusts, and diet 83 included seven controls and six infected locusts (20 controls and 24 infected locusts in total). All 44 libraries were retained in the primary analysis.

2.2 Host, pathogen, and diets

Study animals

Fifth (final) instar female *S. gregaria*, three days post-molt, were obtained from a gregarious colony maintained at Arizona State University from laboratory lines originating at the University of Leicester (United Kingdom) and the University of Haifa-Oranim (Israel). Age was controlled during density rearing to reduce age-dependent transcriptional variation, and the three-day interval allowed cuticular hardening before inoculation. Colony conditions were 34 degrees C during the day, 25 degrees C at night, 20-50% relative humidity, and a 14 h:10 h light:dark cycle. Colony locusts received organic romaine lettuce, wheat grass, and wheat bran. Experimental locusts were housed individually in ventilated plastic containers (18.9 x 13.5 x 9.5 cm) with food and water provided ad libitum.

Fungal culture and inoculation

Locusts were challenged with *M. robertsii* isolate DWR2009 (also designated ARSEF 10343), originally isolated from soil in the United States. Conidia were produced by biphasic solid-state fermentation following Jaronski and Jackson (2012), suspended in Orchex 792 paraffinic oil, and adjusted to 1×10^9 conidia mL⁻¹. Concentration was determined with a hemocytometer and viability was confirmed by germination on potato dextrose agar at 25-26 degrees C. Infected locusts received 1 microliter of the conidial suspension at the hind-leg

234 femur-trochanter junction using a Hamilton Gastight syringe with a repeating dispenser and
 235 blunt-ended PTFE needle. Controls received 1 microliter of sterile Orchex 792 oil and were
 236 handled identically.

237 Artificial diets

238 Chemically defined diets followed Dadd (1961), Simpson and Abisgold (1985), and the
 239 formulation used by Zembruski et al. (2023). Each diet contained 42% macronutrient by dry
 240 mass and was supplemented with Wesson's salts (2.4%), cholesterol (0.5%), linoleic acid (0.5%),
 241 ascorbic acid (0.3%), and a multivitamin mixture (0.2%). Protein was supplied as a 3:1:1 mixture
 242 of casein, peptone, and albumen; carbohydrate was supplied as equal parts sucrose and dextrin.
 243 Diet and water were replenished as needed.

244

245 2.3 Tissue sampling and RNA extraction

246

247 Fat body was collected 96 h after inoculation because this tissue integrates insect
 248 immunity, nutrient storage, and intermediary metabolism. A single experimenter performed all
 249 dissections under sterile, RNase-free conditions. Instruments were flame sterilized and
 250 sequentially treated with 10% bleach, distilled water, 70% ethanol, and RNaseZap before each
 251 dissection. Locusts were surface sterilized with 70% ethanol, and abdominal fat body was
 252 removed while avoiding the gut, trachea, and heart. Tissue was transferred immediately to
 253 nuclease-free tubes, flash-frozen in liquid nitrogen, and stored at -80 degrees C.

254 Total RNA was extracted on a Promega Maxwell RSC instrument with the simplyRNA
 255 Tissue Kit according to the manufacturer's protocol. RNA concentration was measured with a
 256 Qubit 4 Fluorometer and the Qubit RNA HS Assay. RNA integrity and dissection quality were
 257 assessed before library preparation.

258 2.4 Library preparation and sequencing

259 Stranded total-RNA libraries were prepared at Texas A&M University with the Illumina
 260 Stranded Total RNA kit and Ribo-Zero rRNA depletion. Pooled libraries were sequenced by
 261 Novogene on an Illumina NovaSeq X Plus 25B platform as 150-bp paired-end reads, targeting
 262 approximately 45 million read pairs per library.

263 Residual rRNA was quantified from assigned gene-level fragments before filtering. The
 264 median residual rRNA fraction was 7.9% per library (range 3.3-41.7%). All 12,402 gene
 265 identifiers in the project rRNA exclusion list were removed before expression filtering,
 266 normalization, ordination, and differential expression testing.

267 2.5 Read processing, competitive mapping, and gene counting

268 RNA-seq preprocessing was executed on a high-performance computing cluster with a
 269 run-scoped Snakemake workflow. Paired reads were processed with fastp using automatic
 270 paired-end adapter detection, removal of the first two bases of each read, and a minimum
 271 retained length of 50 bp. JSON and HTML quality reports were retained for each library.

272 Trimmed reads were aligned in two-pass mode with STAR v2.7.10b. The primary branch
 273 used a competitive reference comprising the *S. gregaria* assembly GCF_023897955.1 and the *M.*
 274 *robertsii* ARSEF 23 assembly GCF_000187425.2. The latter is a close reference genome rather
 275 than the exact challenge isolate DWR2009/ARSEF 10343. A host-only alignment branch was
 276 retained as a mapping sensitivity analysis. Genome indices used sjdbOverhang = 149 and
 277 alignIntronMax = 2,500,000, selected after examination of *Schistocerca* intron lengths.
 278 Coordinate-sorted BAMs and transcriptome BAMs were retained and indexed with samtools
 279 v1.21.

280 Gene-level fragments were quantified with featureCounts (Subread) using paired-end
 281 fragment counting (--countReadPairs), feature types transcript and exon (-t transcript,exon), and
 282 gene_id as the grouping attribute (-g gene_id). This span-inclusive definition assigns fragments
 283 overlapping either annotated feature class to a gene and can capture signal consistent with
 284 incompletely processed or nascent RNA; an exon-only matrix was analyzed separately as a
 285 count-definition sensitivity check.

286 Sequence placement was assigned from the official assembly metadata rather than scaffold-name
 287 patterns. The primary manuscript analysis was restricted to non-rRNA genes on the main
 288 assembled chromosomes. All-scaffold results were retained as a placement sensitivity analysis,
 289 and competitive-host and host-only placed-chromosome DEG totals were compared to assess
 290 mapping dependence.

291 2.6 Differential expression and sensitivity analyses

292 DESeq2 was applied to raw gene counts after rRNA and sequence-placement filtering.
 293 Genes entered a model when they had at least 10 counts in at least three libraries. Automatic
 294 count replacement was disabled (minReplicatesForReplace = Inf), and Cook's-distance filtering
 295 was disabled for planned contrasts (cooksCutoff = FALSE), so all 44 samples contributed to the
 296 primary estimates. A gene was called differentially expressed at a Benjamini-Hochberg adjusted
 297 P value below 0.05 and an absolute log2 fold change of at least 1. Variance-stabilized counts
 298 were used for PCA and heatmaps, not for DESeq2 testing.

299 A six-level diet-by-treatment group model ($\sim 0 + \text{group}$) generated direct within-diet
 300 infection contrasts and diet contrasts separately among controls and infected locusts. A second
 301 model ($\sim \text{Diet} + \text{Treatment}$) estimated the average infection effect across diets. Diet-dependent
 302 infection responses were evaluated first with an omnibus likelihood-ratio test comparing a full
 303 model ($\sim \text{Diet} + \text{Treatment} + \text{Diet:Treatment}$) with a reduced additive model ($\sim \text{Diet} +$

Treatment), and then with three pairwise difference-of-differences contrasts. Omnibus genes were called at a Benjamini-Hochberg FDR below 0.05; because the LRT tests multiple interaction coefficients jointly, effect size was summarized using the largest absolute pairwise interaction log2 fold change. For diet contrasts, diet 50 served as the reference for comparisons with diets 33 and 83, and diet 33 served as the reference for the 83-versus-33 comparison. Positive log2 fold changes indicate higher expression in the named numerator condition.

Library size, variance-stabilized PCA, sample-distance heatmaps, and heatmaps of highly variable genes were examined for sample-level quality control. Centroids were calculated for each diet-by-treatment group, and lines joining control and infected centroids summarized the direction of the infection shift within each diet.

The primary analysis retained every library. A supplementary sample-retention sensitivity analysis repeated the principal DEG contrasts after removing samples 1007, 1036, 1037, and 1039, which had been identified during exploratory ordination as having atypical global expression profiles. Because these libraries passed sequencing and mapping quality controls and no technical cause was established, this secondary branch was used only to evaluate robustness. Additional sensitivities compared exon-only with transcript-plus-exon counts and main-chromosome with all-scaffold analyses. None of these sensitivity branches replaced the all-sample, transcript-plus-exon, main-chromosome primary analysis.

2.7 Gene classification and functional enrichment

Reference annotation and local crosswalk tables classified genes as protein coding, long non-coding RNA (lncRNA), or other/unknown. Because every listed rRNA locus was removed before modeling, the latter category contains non-rRNA features or genes without a resolved biotype.

Functional descriptions, Gene Ontology (GO) terms, and KEGG orthology/pathway assignments were obtained from the project eggNOG annotation and linked to gene identifiers through CDS records in the reference GFF. KEGG results were shown in a eukaryote-inclusive view and were interpreted as orthology-based, hypothesis-generating annotations rather than direct evidence that an insect pathway was activated.

Over-representation tests were performed with clusterProfiler::enricher using the 14,132 expressed, non-rRNA main-chromosome genes entering the primary DESeq2 analysis as the background. P values were adjusted by the Benjamini-Hochberg method. For the two expression clusters, enrichment foregrounds were restricted to genes differentially expressed in at least one within-diet infected-versus-control contrast. GO Biological Process, Cellular Component, and Molecular Function terms and eukaryote-compatible KEGG pathways were summarized separately. Term-to-gene reverse-index tables were retained so each enrichment result could be traced to its contributing genes.

Genes significant in any primary contrast were clustered from row-standardized variance-stabilized expression using Euclidean distance and complete-linkage hierarchical clustering. Candidate cuts from $k = 2$ to $k = 10$ were compared by silhouette width; $k = 2$ had the highest

mean silhouette width and defined the two primary expression programs. Genes were ordered within each cluster without averaging individual-gene expression, preserving sample-level heterogeneity in the heatmap. Cluster enrichment used within-diet infection DEGs as foregrounds and the expressed, rRNA-filtered main-chromosome genes as the universe.

2.8 Curated physiological candidate genes

A post hoc curated candidate list was used to describe immune recognition and signaling, humoral immunity, cellular immunity, detoxification, protein synthesis and translation, nutrient sensing and TOR/insulin signaling, nitrogen and amino-acid metabolism, and storage hexamerins. The primary placed-chromosome analysis retained 99 candidates. Genes had originally entered this list when they were significant in at least one examined comparison; consequently, the displayed proportions were treated as descriptive summaries rather than independent category-level hypothesis tests. For each contrast, candidates were classified as upregulated, downregulated, or not significant using the same DEG thresholds as the genome-wide analysis. The all-scaffold candidate set was retained as a placement sensitivity analysis.

2.9 Phase-reference specificity analysis

Because experimental locusts were housed individually for 96 h, the union of fat-body genes significant in at least one within-diet infection contrast was compared with independent solitarious-versus-gregarious DEG sets from *S. gregaria* head and thorax. Gene identifiers were matched directly and set intersections were summarized without re-estimating expression across tissues. These reference data came from different tissues and experimental contexts; the comparison was therefore used only to assess whether the dominant infection response broadly recapitulated known density-associated expression, not to infer a phase change in the experimental fat body.

2.10 Fungal alignment proxy

Uniquely aligned fragments assigned to the fungal component of the competitive reference were summarized as counts and as a percentage of uniquely aligned competitive-reference fragments. Because the mapping reference was not strain matched and fungal load was not confirmed post mortem by qPCR or culture, this quantity was treated only as RNA-seq evidence of fungal transcriptional signal. A parallel qPCR validation would have been preferable, but no matching tissue remained for that assay.

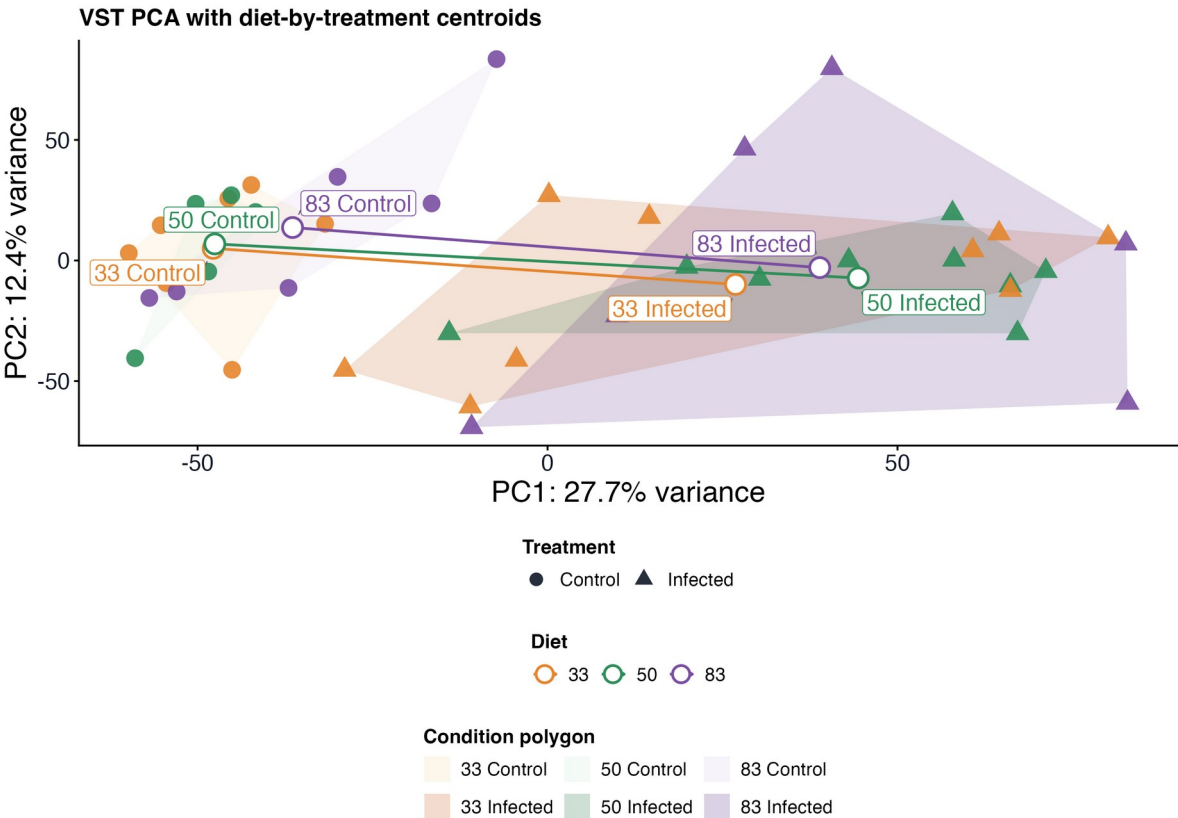
3. Results

3.1 Fungal infection dominates fat body expression across diets

The first two principal components of variance-stabilized expression explained 27.7% and 12.4% of total variance, respectively (Fig. 1). Control samples occupied the negative portion of PC1, whereas most infected samples shifted toward positive PC1 values. The control-to-infected

380 centroid moved in the same general direction for each diet, although the hulls overlapped and
381 within-group dispersion differed, particularly among infected locusts. Infection therefore
382 represented the dominant common axis of variation, with diet and inter-individual heterogeneity
383 modifying the position and breadth of each group.

384 Fungal alignment signal was low in most fat-body libraries but highly heterogeneous
385 among infected individuals. Median fungal percentages were 0.00138%, 0.00097%, and
386 0.00084% in infected diets 33, 50, and 83, respectively, while a small number of infected
387 libraries reached 0.64-3.13%. Controls remained near background. Because these values came
388 from competitive RNA-seq alignment rather than a strain-matched load assay, they were used to
389 describe detectable fungal RNA, not to classify infection success.



390 **Figure 1.** Fat body transcriptomes shift with fungal infection across the three diets. PCA used variance-stabilized counts
391 from expressed, non-rRNA genes on the main *S. gregaria* chromosomes. Points represent individual libraries, shapes
392 indicate treatment, colors indicate diet, shaded polygons show the observed breadth of each diet-by-treatment group, open
393 circles mark centroids, and lines connect the control and infected centroids within each diet.

395 3.2 Differential expression across infection and diet contrasts

396 The average infected-versus-control contrast identified 1,691 DEGs across diets, including 893
397 genes higher in infected locusts and 798 higher in controls (Fig. 2). Within diets, infection
398 altered 1,016 genes in diet 33 (552 higher in infected and 464 higher in controls), 2,009 genes in
399 diet 50 (1,263 and 746), and 1,030 genes in diet 83 (658 and 372). Protein-coding genes formed

400 the majority of each response, but lncRNAs contributed 51, 132, and 62 DEGs in diets 33, 50,
 401 and 83, respectively.

402 Direct diet effects were much smaller among controls: pairwise contrasts yielded 15
 403 DEGs between diets 33 and 50, 60 between diets 83 and 33, and 59 between diets 83 and 50.
 404 Diet contrasts among infected locusts yielded 303, 86, and 59 DEGs for the same comparisons.
 405 The particularly large infected 33-versus-50 contrast was strongly directional, with 290 genes
 406 higher in diet 50 and 13 higher in diet 33. Across the three within-diet infection responses, 467
 407 genes were shared by all diets, whereas 211, 1,069, and 320 were unique to diets 33, 50, and 83,
 408 respectively.

409 Formal interaction testing indicated that most of the infection response was shared rather than
 410 broadly diet specific. The omnibus diet-by-infection LRT identified nine genes in the primary
 411 transcript-plus-exon main-chromosome analysis; all nine had a maximum pairwise interaction
 412 effect of at least $|\log_2\text{FC}| = 1$. Pairwise difference-of-differences contrasts identified 0, 58, and
 413 11 DEGs for diets 33 versus 50, 83 versus 50, and 83 versus 33, respectively, and all nine
 414 omnibus genes appeared in at least one of the two contrasts involving diet 83. Under exon-only
 415 counting, no gene passed the omnibus interaction FDR. The evidence for diet-dependent
 416 modification was therefore selective and sensitive to count definition, despite the marked
 417 differences in within-diet DEG totals.

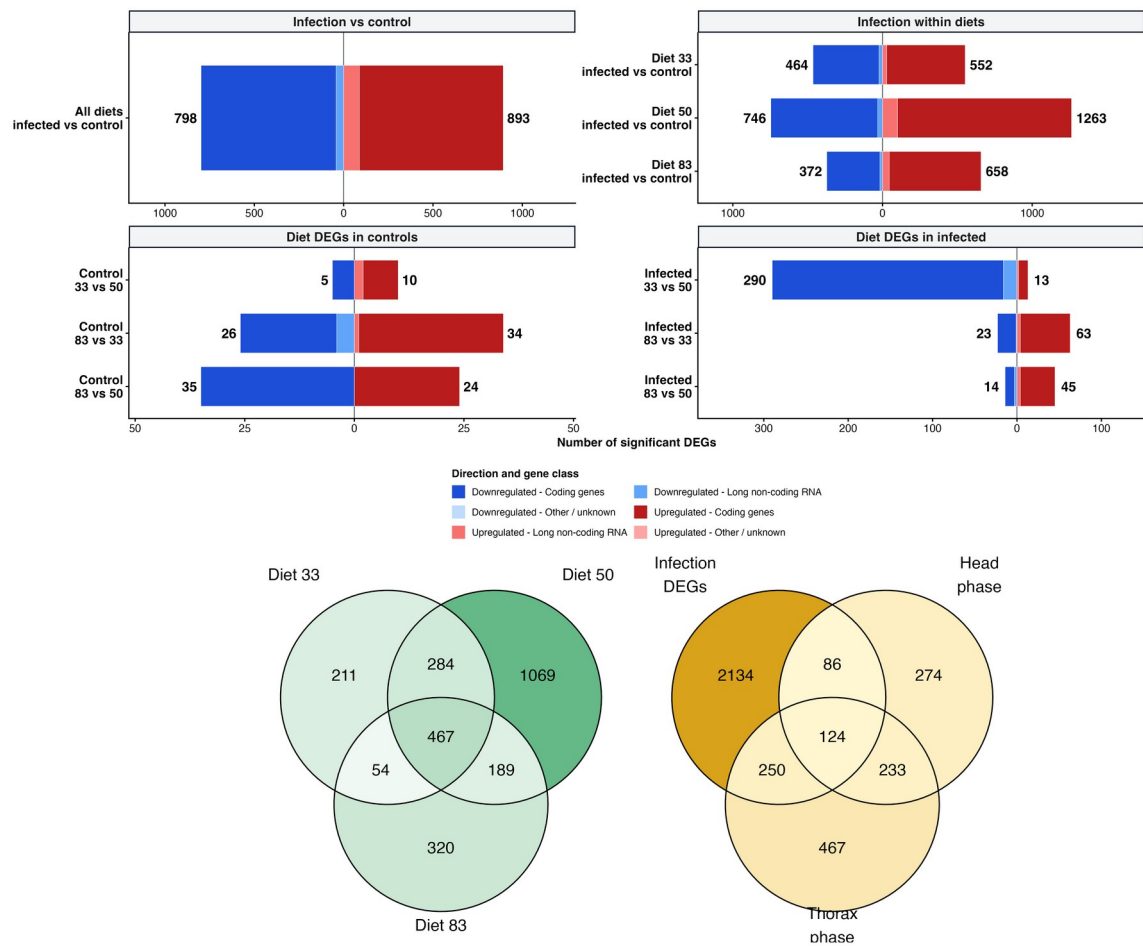


Figure 2. Differential-expression burden and overlap in the all-sample, main-chromosome analysis. The upper panels show coding, lncRNA, and other/unknown DEGs by direction for the average infection response, infection within each diet, diet contrasts among controls, and diet contrasts among infected locusts. Red values are higher in the named numerator condition and blue values are higher in the denominator. The lower-left Venn diagram partitions the three within-diet infection DEG sets. The lower-right diagram compares their union with external head and thorax solitary-versus-gregarious DEG sets; this overlap provides phase-related specificity context but does not establish phase effects in fat body. Of 2,594 infection-responsive genes in this union, 2,134 did not overlap either reference phase set. Annotated rRNA genes were absent from every set.

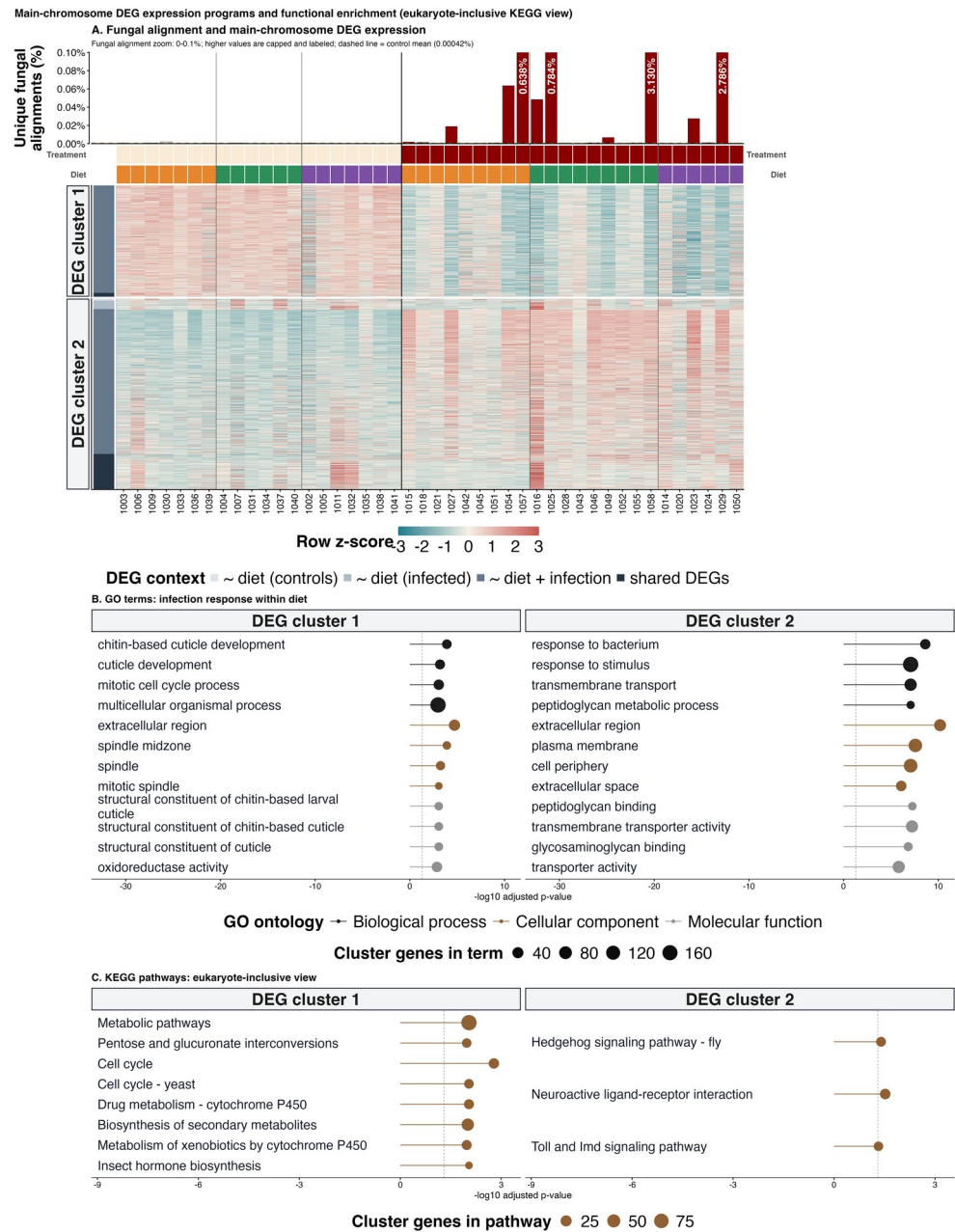


Figure 3. Main-chromosome DEG expression programs and functional enrichment. (A) Gene-level row-z-scored variance-stabilized expression for 2,949 DEGs, separated by complete-linkage Euclidean clustering into the two-cluster cut with the strongest mean silhouette width and ordered with controls before infected samples. The upper bar shows the percentage of unique competitive alignments assigned to *M. robertsii*; values above the 0.1% display range are labeled. (B) GO terms enriched among within-diet infection DEGs in each cluster. (C) Eukaryote-inclusive KEGG pathways enriched in the same foregrounds. Point size indicates the number of cluster genes in a term. Cluster-1 enrichment values are mirrored to the left only to separate the panels; the sign does not represent regulation direction.

3.3 Infection-associated expression programs and functional enrichment

The union of genes significant in the primary contrast set comprised 2,949 main-chromosome genes. Complete-linkage clustering of Euclidean expression distances resolved two broad programs without averaging gene-level expression (Fig. 3); the two-cluster cut had the

strongest mean silhouette width among candidate cuts from $k = 2$ to $k = 10$. Cluster 1 contained 1,086 genes and generally showed higher expression in controls and lower expression after infection. Cluster 2 contained 1,863 genes and showed the reciprocal pattern. Most genes in both clusters were associated with an infection-within-diet contrast; smaller subsets reflected diet effects or membership in more than one contrast context.

Among within-diet infection DEGs, cluster 1 was enriched for chitin-based cuticle development, mitotic cell-cycle and spindle functions, oxidoreductase activity, broad metabolism, cytochrome P450-associated xenobiotic metabolism, and insect hormone biosynthesis. Cluster 2 was enriched for response to bacteria and stimulus, extracellular and plasma-membrane compartments, peptidoglycan metabolism and binding, transmembrane transport, and Toll and Imd signaling. These reciprocal programs are consistent with reduced investment in cuticle, cell-cycle, and selected metabolic processes alongside increased expression of extracellular defense and recognition functions during infection.

Only 325 cluster-1 and 396 cluster-2 foreground genes had GO annotations, and 264 and 318, respectively, had KEGG assignments. The enrichment results therefore summarize the annotated subset rather than every DEG. Fungal alignment percentages displayed above the heatmap showed that the host expression programs occurred across a wide range of fungal RNA signals, but this visual correspondence was not interpreted as a validated relationship with pathogen load.

3.4 Curated immune and physiological candidate genes

The main-chromosome curated analysis retained 99 genes across eight immune and protein-anabolism categories (Fig. 4). Within each diet, infection increased a substantial fraction of genes involved in innate immune recognition and signaling and humoral immunity. Responses in cellular immunity and detoxification were more variable, and several genes in these small categories were not significant.

Protein-anabolism categories contained both increased and decreased responses. Protein synthesis and translation included a small upregulated fraction and a larger downregulated fraction in every diet. Nutrient-sensing and nitrogen/amino-acid categories tended to contain more upregulated genes in diet 50 than in diets 33 or 83, whereas storage hexamerins were mostly downregulated in diet 50 and not significant in diets 33 and 83.

Across-diet contrasts among infected locusts contained relatively few significant curated genes. Because the curated list had been assembled from genes significant in at least one comparison, its status proportions were not subjected to an independent category-level significance test. The figure is therefore a descriptive map of infection-responsive physiological candidates and identifies categories for future testing with an annotation-defined, a priori gene set.

Restricting the curated list to main chromosomes reduced it from 163 all-scaffold candidates to 99. This placement filter was applied consistently to the primary PCA, differential-expression, clustering, enrichment, and curated analyses; the all-scaffold results are retained as a supplementary sensitivity.

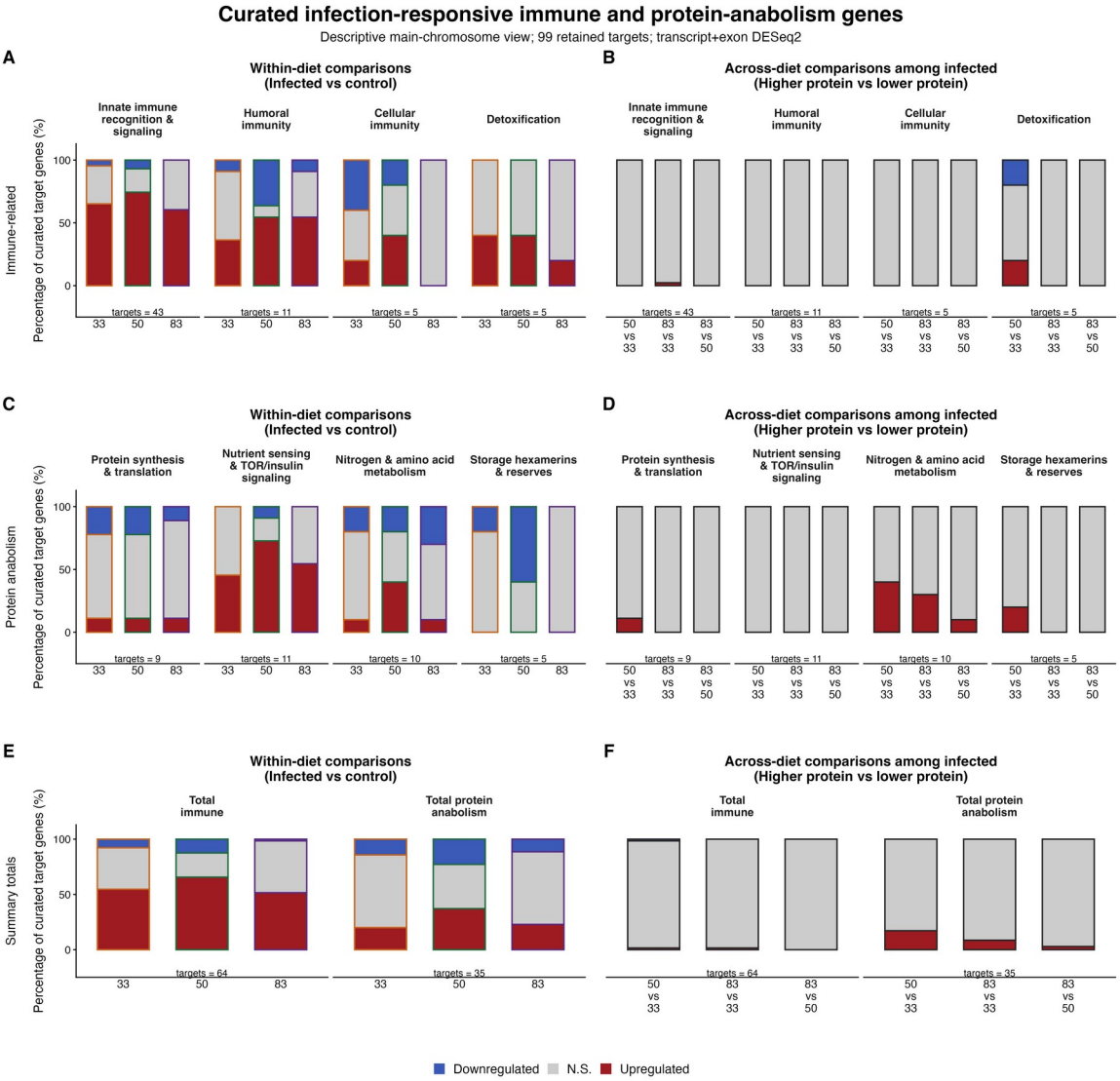


Figure 4. Curated immune and protein-anabolism candidate genes on the main *S. gregaria* chromosomes. Stacked bars show the percentage of genes that were upregulated, downregulated, or not significant. (A, C) Infection responses within each diet. (B, D) Pairwise diet contrasts among infected locusts, oriented as higher- versus lower-protein diets. The number below each functional heading is the number of retained candidate genes. Candidate membership was selected post hoc from genes significant in at least one contrast, so this panel is descriptive and does not present independent category-level p-values.

4. Discussion

Metarhizium infection was the principal driver of transcriptional variation in the desert locust fat body. Control-to-infected centroids moved in a common direction across diets, the average infection contrast identified 1,691 DEGs, and 467 genes were shared by all three within-diet infection responses. The balanced diet produced the largest descriptive DEG total, but the

omnibus interaction identified only nine genes under the primary transcript-plus-exon definition and none under exon-only counting. Different numbers of significant genes among diets therefore should not be interpreted as evidence of broad diet-dependent interaction. Instead, the data support an extensive shared infection programme with selective and count-definition-sensitive dietary modification.

The two expression programs provide a physiological summary of this common response. One programme decreased during infection and was associated with cuticle, cell-cycle, and selected metabolic processes, while the other increased and was associated with extracellular defense, peptidoglycan recognition, transport, and Toll/Imd signaling. Infection therefore involved both immune activation and broad reallocation away from structural and proliferative processes. The higher mean silhouette width of the two-cluster cut supports this broad division, while the gene-level heatmap also makes clear that individual genes and locusts varied within each programme.

The immune-investment hypothesis predicted stronger or broader immune transcription under the carbohydrate-biased diet. Instead, recognition, signaling, and humoral-defense candidates were commonly induced within every diet, and cluster enrichment identified extracellular defense and Toll/Imd-associated functions as general features of infection. The curated list was assembled post hoc from genes significant in at least one contrast, so its category proportions are descriptive rather than independent tests. Even with that limitation, the observed pattern does not support generalized enhancement of immune transcription as the explanation for improved survival under carbohydrate-biased nutrition.

Earlier physiological and metabolomic work in this system showed that higher-protein diets increased hemolymph protein and branched-chain amino-acid availability and were associated with accelerated mortality and fungal development, despite elevated phenoloxidase activity (Tahir et al., 2026; Tahir et al., submitted). The present mixed responses in nutrient sensing, amino-acid metabolism, translation, and storage genes are compatible with infection-driven metabolic reorganization, but they do not establish that diet 50 or diet 83 produced a uniformly more anabolic fat body. Together, the physiological and transcriptional results are more consistent with selective differences in host metabolic state, resource availability, or tolerance than with simple immune failure, while leaving the causal mechanism unresolved.

Fungal RNA provided contextual evidence of among-individual heterogeneity. Most infected fat-body libraries contained little uniquely aligned fungal signal, whereas several individuals contained markedly higher values. This heterogeneity could reflect differences in infection progression, fungal localization, or sampling of a tissue in which fungal biomass was low at 96 h. The competitive reference used ARSEF 23 rather than the challenge isolate DWR2009/ARSEF 10343, and no matching tissue remained for qPCR or culture. Consequently, these alignments were not interpreted as validated fungal load, used to reclassify infected individuals, or assumed to explain host-expression variability without a formal association test.

The phase-reference comparison addressed whether 96 h of individual housing broadly recapitulated known density-associated expression. Of 2,594 fat-body infection DEGs in the set comparison, 2,134 did not overlap either the head or thorax phase-reference set. The smaller overlap may identify genes involved in cross-tissue physiological plasticity or shared stress responses, but the reference data came from different tissues and experimental contexts. It therefore does not demonstrate a phase transition in the experimental fat body or attribute the infection response to solitarization.

Sequence placement was another important source of uncertainty. A large historical DEG signal occurred on unplaced scaffolds, especially in the infected diet-33-versus-diet-50 contrast. Taxonomic, cross-genome, and homology audits did not justify treating every unplaced locus as contamination, but their provenance was less secure than that of genes on assembled chromosomes. Restricting the primary analysis to main chromosomes provides a conservative host interpretation, while preserving all-scaffold results for sensitivity and future investigation rather than deleting those loci. The major interpretation of broad immune induction and heterogeneous protein-anabolism responses was retained when all scaffolds were shown, although the curated denominator increased from 99 to 163 genes.

Functional enrichment in a non-model insect remains constrained by annotation coverage and orthology. Only a subset of genes in either expression cluster carried GO or KEGG assignments, and broad KEGG pathway names may reflect conserved orthologous functions rather than a locust-specific pathway definition. The use of transcript-plus-exon counts increased sensitivity relative to exon-only counting and can include signal consistent with nascent or incompletely processed RNA. The principal infection response remained extensive under exon-only counting, but the absence of an exon-only omnibus interaction makes the diet-dependent component less robust than the shared infection response. Count-definition, sequence-placement, and sample-retention results are therefore reported explicitly, and enrichment remains hypothesis-generating until supported by protein, metabolite, or functional assays.

Together, these results support a model in which *M. robertsii* induces a conserved and extensive fat body response across nutritional environments, while dietary macronutrient balance modifies a comparatively small and method-sensitive subset of responses. Canonical immune functions were induced across diets, whereas metabolic, translation, nutrient-sensing, and storage-related genes responded heterogeneously. The carbohydrate-biased diet's survival advantage is therefore unlikely to reflect generalized enhancement of immune transcription at 96 h. Future experiments combining matched fungal-load qPCR, proteomics, metabolite flux, and pathway-specific perturbation will be required to resolve whether host resource allocation, pathogen nutrition, or tolerance drives the observed survival differences.

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 573 O’odham and Piipaash peoples whose land stewardship we benefit from daily.

574 **Data and code availability**

575 All scripts and code used for the fat body transcriptomes processing, mapping and subsequent
 576 differential gene expression analysis are available and curated here [https://songlab-](https://songlab-genomics.github.io/gregaria-diet-infection-interaction/)
 577 [genomics.github.io/gregaria-diet-infection-interaction/](https://songlab-genomics.github.io/gregaria-diet-infection-interaction/)

578

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583

584 **Author Contributions**

585 Author contributions following the CRediT taxonomy

586 SMT: Conceptualization, Methodology, Formal analysis, Investigation, Data Curation, Writing –
 587 original draft preparation, Visualization, Revision

588 MT: Methodology, Formal analysis, Investigation, Data Curation, Writing – original draft
 589 preparation, Visualization, Revision

590 EB: Methodology, Formal analysis, Investigation, Data Curation, Visualization, Revision

591 TN: Assisted with Methodology, Investigation

592 SN: Assisted with Methodology, Investigation

593 NR: Assisted with Methodology, Investigation, Formal analysis

594 SM: Assisted with Formal analysis

595 JL: Assisted with Methodology, Investigation

596 SJ: Methodology, Resources

597 HS: Resources, Methodology, provided feedback on Formal analysis and visualization, Writing -
 598 review and editing, Supervision

599 JFH: Conceptualization, Resources, Methodology, provided feedback on Formal analysis and
 600 visualization, Writing - review and editing, Supervision
 601 AJC: Conceptualization, Resources, Methodology, Formal analysis, Visualization, Writing -
 602 original draft preparation, Supervision, Funding acquisition

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Supplementary Information for:

**Fat body transcriptional responses to dietary macronutrient balance and *Metarhizium*
 infection for desert locusts, *Schistocerca gregaria***

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Figure S5. Title. Description.

Figure S1. Outlier-removal sensitivity analysis
All comparisons use the rRNA-filtered count matrix and the same DEG thresholds as the main figures

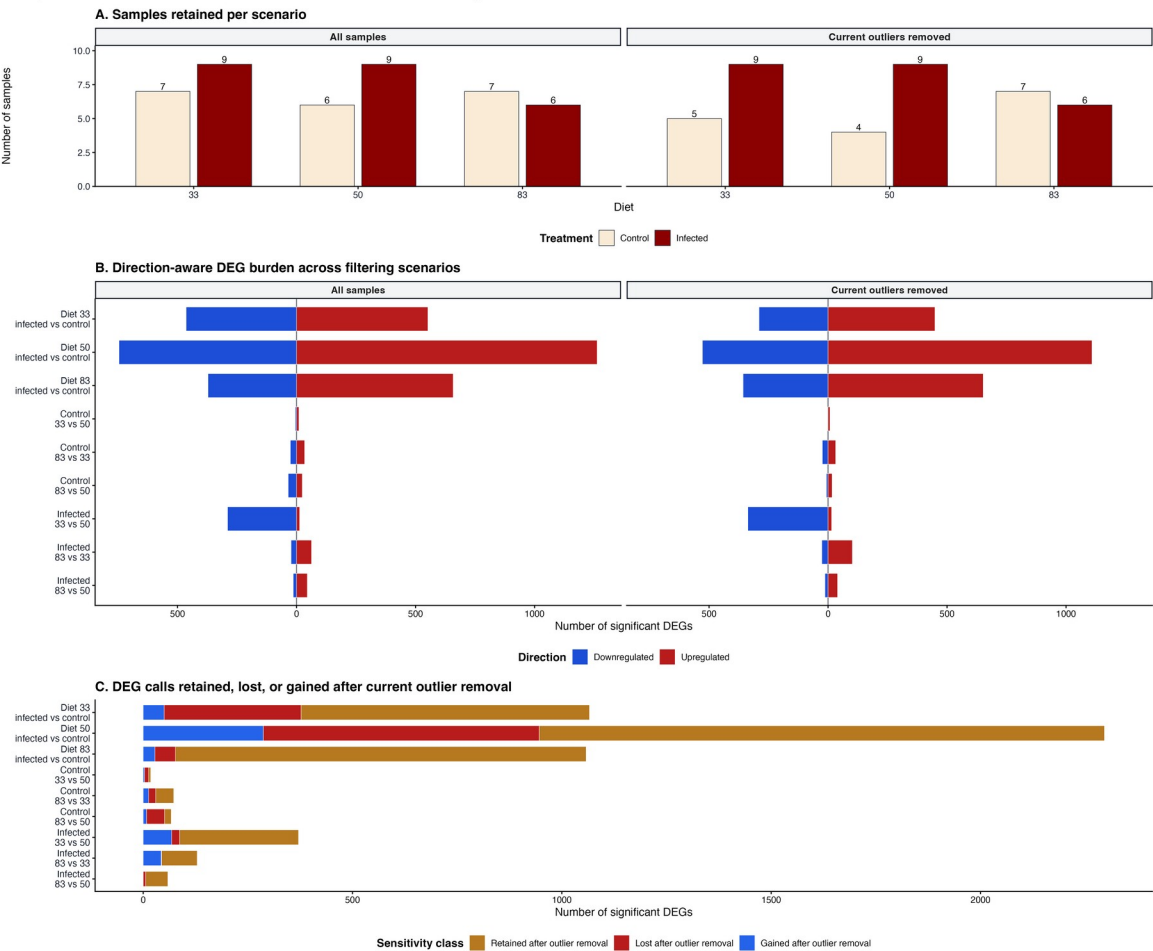


Figure S1. Sample-retention sensitivity analysis. Primary DEG totals from all 44 libraries are compared with a secondary analysis excluding samples 1007, 1036, 1037, and 1039. This analysis evaluates the influence of atypical libraries; no sample was excluded from the primary results.

Figure S2. Chromosomal localization of DEG contexts

Primary paper DEG sets contain only official nuclear-chromosome genes

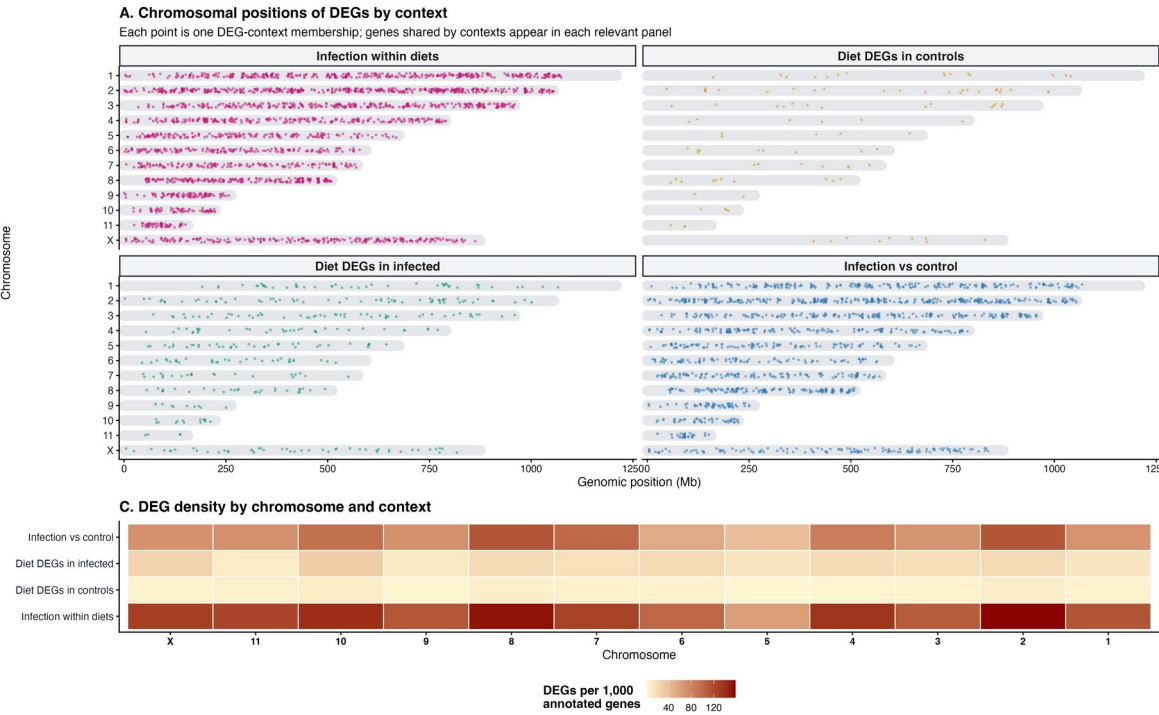


Figure S2. Chromosomal localization of differential expression. Main-chromosome DEGs are shown by genomic position and contrast context, providing a genome-wide view of whether signals are broadly distributed or concentrated in particular chromosome regions.

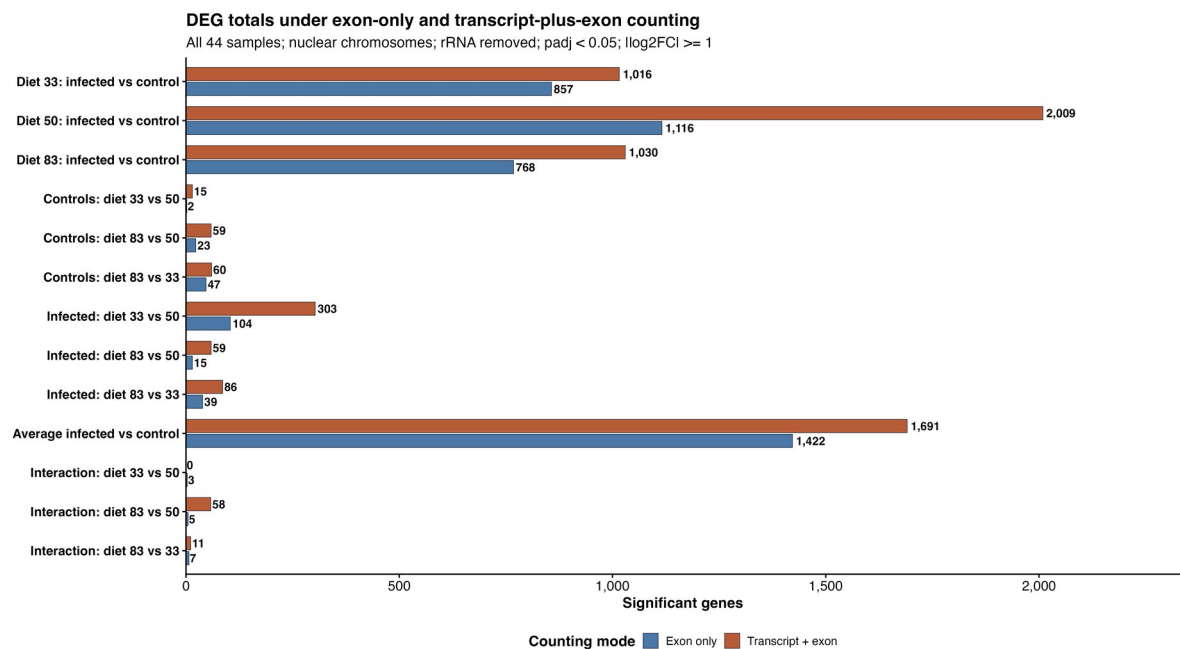


Figure S3. Count-definition sensitivity. Numbers of significant genes obtained with exon-only counts are compared with counts assigned from transcript and exon features and grouped by gene_id. Both matrices were depleted of annotated rRNA and restricted to the same primary chromosome scope.

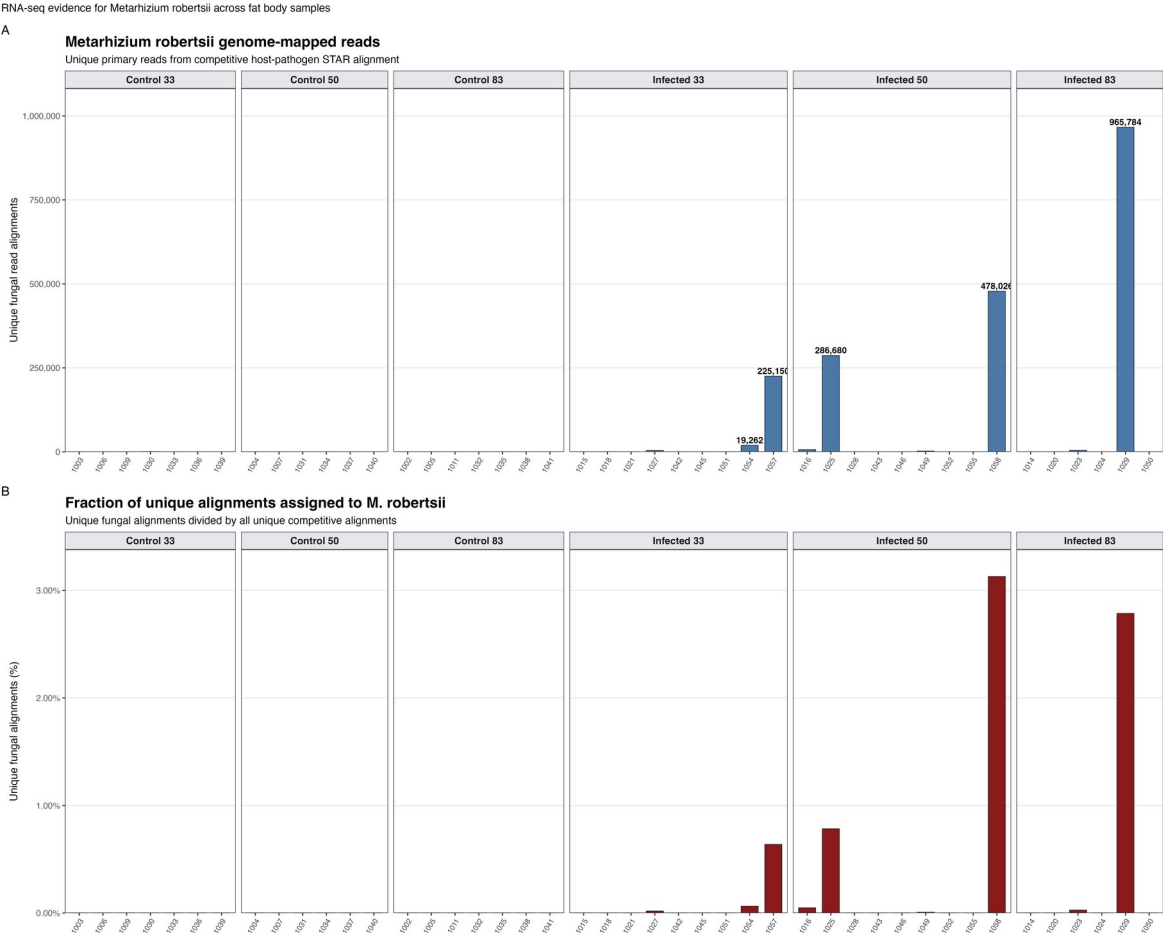


Figure S4. Competitive-alignment evidence for fungal RNA. Full-range and zoomed views show uniquely aligned *M. robertsii* fragments and their percentage of unique competitive-reference alignments. Horizontal control means provide a background reference. These values are RNA-seq proxies, not qPCR-validated fungal loads.

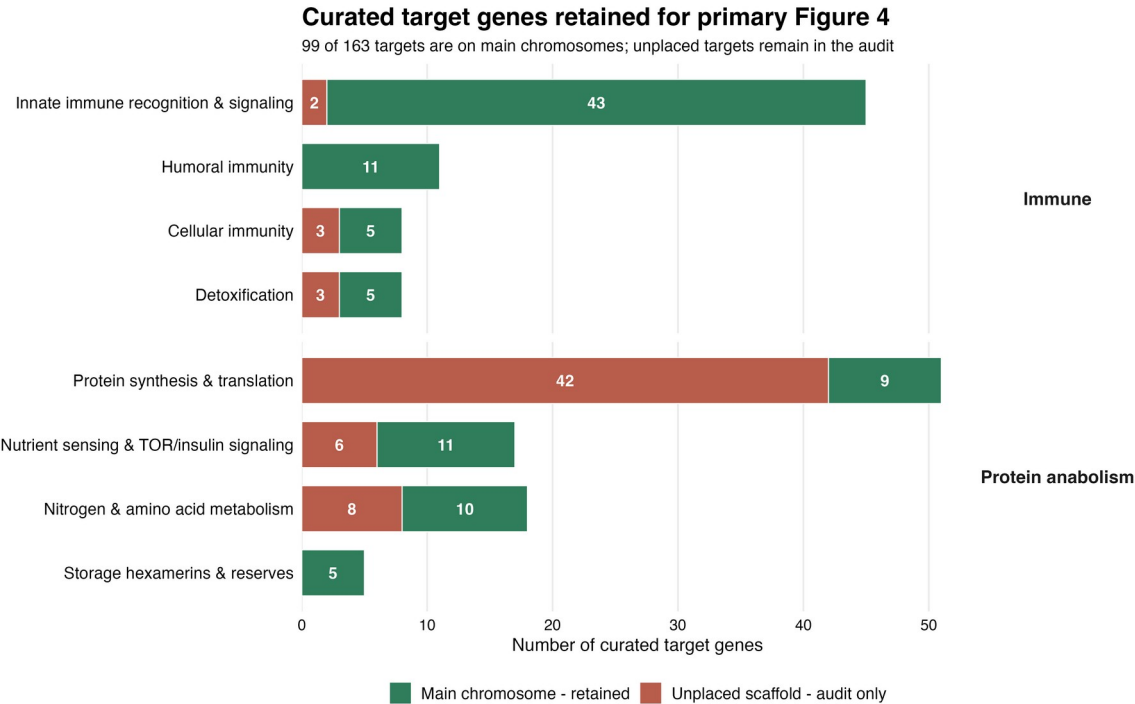


Figure S5. Sequence-placement sensitivity for the curated candidate analysis. Results from all annotated scaffolds are compared with the primary main-chromosome subset, showing which changes arise from excluding unplaced or otherwise non-primary assembly sequences.

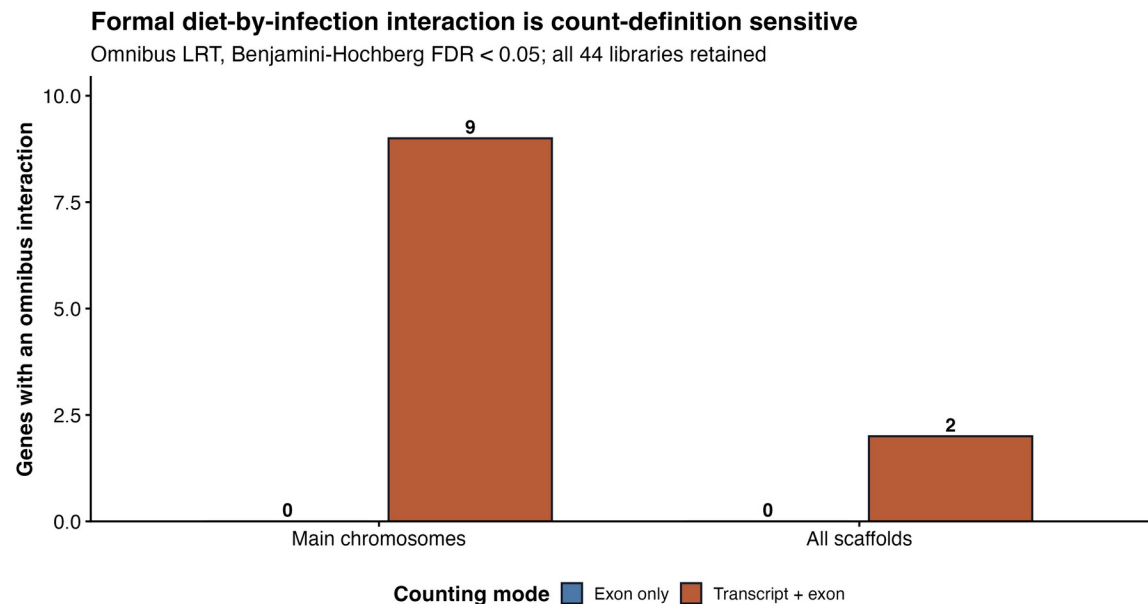


Figure S6. Formal diet-by-infection interaction sensitivity. Omnibus likelihood-ratio tests compare the full diet-plus-infection-plus-interaction model with the reduced additive model under exon-only and transcript-plus-exon counting. All 44 libraries are retained and annotated rRNA genes are excluded.

Draft cover letter

Greetings Editors,

On behalf of my co-authors, I am pleased to submit our manuscript entitled “**Fat body transcriptional responses to dietary macronutrient balance and *Metarhizium* infection for desert locusts, *Schistocerca gregaria***” for consideration for publication in *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology*.

Our study addresses the critical question of how dietary protein-to-carbohydrate (p:c) ratios affect immune function and pathogen susceptibility in locusts exposed to the entomopathogenic fungus *Metarhizium robertsii*. Contrary to the common assumption that heightened immune activation necessarily improves survival, this study builds on two earlier studies to show that this hypothesis is not supported and that nutrient availability for the fungus is likely more important. Tahir et al. (2026) demonstrated a diet-dependent trade-off: locusts consuming a protein-biased diet showed increased phenoloxidase and prophenoloxidase activity but suffered higher mortality and greater fungal sporulation following infection. Tahir et al. (in review) tested the hypothesis that the protein-biased diets provide the fungus with nutritionally favorable hemolymph that supports fast growth. The results were consistent with that hypothesis, with protein-biased diets elevating protein and branch-chained amino acids critical for fungal growth and infection decreasing levels. Due to the significant diet x infection interaction, total protein may be most important. We are submitting a copy of the paper under review for reference.

Our study complements earlier physiological measurements by testing the interactive effects of diet and *Metarhizium* infection on fat body transcription. Infection produced a large conserved response, with a 467-gene core shared among all diets. Although within-diet DEG totals differed, the omnibus interaction identified only nine genes under transcript-plus-exon counting and none under exon-only counting. Immune functions were broadly induced, whereas metabolic and protein-anabolism candidates responded heterogeneously. Together with the earlier hemolymph and survival data, these results argue against generalized enhancement of immune transcription as the carbohydrate-associated survival mechanism and instead motivate tests of host resource allocation, pathogen nutrition, and disease tolerance.

We believe this study will be of interest to *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology*’ readership, particularly because it complements earlier physiological studies to test specific hypotheses about the molecular and biochemical mechanisms underlying organismal response to nutrition and infection. It should also be of broad interest for host-pathogen interaction researchers. The manuscript has not been submitted elsewhere, and all authors have approved its submission.

Thank you for considering our work. We look forward to the possibility of this work being published in *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology*.

Thank you,

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